

LOGIC

Def:-

→ "logic is the study of the methods and principles used to distinguish correct from incorrect reasoning" said by I.M. Copi.

→ "logic is a science of reasoning. The aim of logic is to provide methods, techniques and devices which help in differentiating right reasoning from wrong reasoning and good reasoning from bad reasoning," said by Krishna Jain.

Expi:-

The term **LOGIC** came from Greek word "**LOGOS**" which is sometimes translated as sentence, discourse, reasons, rule and ratio.

In the west, Logic was established as a formal discipline. "Aristotle," who gave it a fundamental place in philosophy. In the east, logic was developed by "Buddhists".

Examples:-

- 1) All squares are rectangles.
- 2) All dogs have good sense of smell.
- 3) A ~~xxx~~ polygon having 3 sides, is a triangle.

Types of logic:-

i) Formal language:-

Formal language is also called deduction.

Formal logic is what we think of as traditional logic or philosophical logic. It is mainly concern with formal systems of logic. A formal system is also called "logic calculus" is used to drive conclusions from one or more premises.

Example:-

P₁ = All men are mortal

P₂ = Ali is a man.

Conclusion:- So, Ali is mortal.

ii) Informal Logic:-

Informal logic is also called Inductive reasoning or Induction.

"The term informal logic is often used to mean the same thing as critical thinking". Sometimes, it is used to refer to the study of reasoning and fallacies in the context of everyday life.

Informal logic studies natural language arguments.

Example:-

P₁ = If you don't study, you'll fail.

$P_2 =$ Ali didn't study

Conclusions - Ali will fail.

QR-II

BS- Physics 2nd semester

→ logic...

→ Importance of logic:-

i) Inductive Reasoning:-

Inductive Reasoning is the process of reaching a general conclusion by examining specific examples.

The conclusion formed by using inductive reasoning is often called **Conjecture** which may or may not be correct.

Inductive reasoning is also called "Inductive logic" or "bottom-up reasoning".

Examples:-

Ex#01:-

Specific observations:-

Baby haider said his first word at the age of 12 month.

Pattern recognition:-

All observed babies says there first word at the age of 12 month.

General conclusion:-

All babies says there first word at the age of 12 month.

Ex#02

Specific observation:-

Parrot is a bird which can fly.

Sparrow is a bird which can fly.

Crow is a bird which can fly.

Pattern recognition:-

All observed birds can fly.

Conclusion:-

All birds can fly.

ii) Deductive Reasoning:-

Deductive Reasoning is the process of reaching a conclusion by applying general principles, assumptions and procedures.

Example:-

1) P_1 Good Students pass exams.

P_2 Ali is a good student.

Result:- Therefore, Ali will pass this exam.

2) P_1 All birds can fly.

P_2 Kivi is a bird.

Result:- Therefore, Kivi can also fly.

3) P_1 What thinks must exist.

P_2 I'm thinking.

Result:- Therefore, I exist

Exercise Use deductive reasoning to show that the following procedure produces a number that

Q:1

is 4 times as the original no.

Procedure:-

For ex:-

Pick a number 6.

The number is 6

x by the no. by 8.

$$6 \times 8 = 48$$

add 6 to the product.

$$6 + 48 = 54$$

divide the sum by 2.

$$54/2 = 27$$

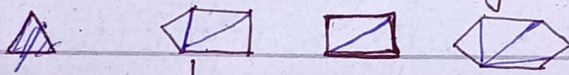
Subtracting 3.

$$27$$

$$24 = 6(4) \text{ Ans.}$$

Q:2 Use inductive reasoning:-

①



~~original~~ Pentagon has 5 sides: 3 triangles.

Quadrilateral has 4 sides: 2 triangles

Hexagon has 6 sides: 4 triangles

② The Series of no. is

4 8 16 32 64 ...??

What is the next number in the series?

Next no. is 128.

Each no. is obtained by multiplying the previous no. by 2.

③ 2, 5, 10, 17 ... ???

Next no. is 26.

Reason Adding consecutive odd numbers.

$$2 + 3 = 5$$

$$5 + 5 = 10$$

$$10 + 7 = 17$$

$$17 + 9 = 26$$

iii) Abductive Reasoning:-

Abductive reasoning or logic is also known as abduction or interference to the best explanation, is a form of reasoning where hypothesis (are) explanation is inferred based on available evidence of observations.

Exp:-

It involves making an educated guess or plausible inference to explain a particular phenomena or set of facts. In abductive logic the goal is to find the most likely or best explanation for the observed data, even if it may not be the only possible explanation or completely ~~for~~ certain.

- Abductive reasoning is formulated by **Charles Sanders Peirce**.
- It starts with incomplete observations and ends with the best predictions which may be true.
- It leads to discoveries and best guess.

Examples:

When a patient presents with certain symptoms. Doctors used abductive reasoning to generate potential explanations or diagnosis. They consider the available evidence (symptoms, medical history and test results) and make an educated guess.

#02 Crime investigation.....

#03 Scientific research....

#04 Marketing and consumer Behaviour....

Explain....

"Proposition"

It is a statement that may be true or false

for example:-

Sky is blue T

$2 + 3 = 8$ F

Men is mortal T

Birds fly in the air. T

Logical operations:-

- | | | | | |
|----------------|----------|-----|--------------|---------|
| 1- Negation | $\sim$ | not | $\sim p$ | not p |
| 2- disjunction | $\vee$ | or | $p \vee q$ | p or q |
| 3- Conjunction | $\wedge$ | and | $p \wedge q$ | p and q |

4- Conditional $\rightarrow/\Rightarrow$ Implies $P \rightarrow Q$ P implies Q
(Implication) if... then... if, P then Q

5- Biconditional $\leftrightarrow$ equivalent $P \leftrightarrow Q$ P if and only if Q
OR P is equivalent to Q.

Definitions:-

1- Negation:-

Let P be any proposition, the negation of P denoted by $\sim P$ or $\sim Q$, is a propositional which is "not P".

Example:-

→ Let P be the proposition:-

As $7+4=11$ then $\sim P$ is $7+4 \neq 11$.

→ Let P be the proposition:-

"the birds fly" then $\sim P$ is "the birds do not fly".

2- Conjunction:-

Let P and Q be two propositions.

The conjunction of P and Q, denoted by $(P \wedge Q)$ is a proposition, which is true when both P and Q are true and false otherwise.

3- Disjunction:-

Let P and Q be two propositions.

The disjunction of P and Q, denoted

Date: _____

Day: _____

by $(P \vee q)$ is a proposition which is false when both P and q are false and true otherwise.

4: Implication or Conditional:-

Let P and q be two propositions. The implication or conditional of P and q , denoted by $P \rightarrow q$, is a proposition which is true false when P is true and q is false and True otherwise. In this form, P is called "antecedent" or "hypothesis" and q is called "consequent" or "conclusion."

5- Equivalence or Bi-Conditional:-

Let P and q be two propositions. The equivalence or Bi-Conditional of P and q , denoted by $(P \leftrightarrow q)$ is a proposition which is true when both P and q have same truth values and false otherwise.

6- Exclusive or:-

Let P and q be two propositions. The Exclusive or of P and q , denoted by $(P \oplus q)$ is a proposition which is True when exactly one of P and q ,

Date: _____

Day: _____

is true and false otherwise.

1	2	3	4	5	6	7	8	9
P	Q	$\sim P$	$\sim Q$	$P \wedge Q$	$P \vee Q$	$P \rightarrow Q$	$P \leftrightarrow Q$	$P \oplus Q$
T	T	F	F	T	T	T	T	F
T	F	F	T	F	T	F	F	T
F	T	T	F	F	T	T	F	T
F	F	T	T	F	F	T	T	F

→ Truth Table:-

"A table that shows the truth values of a proposition is called a truth table."

Truth value of proposition is denoted by T if it is True while Truth value of a proposition is denoted by F if it is false.

Example:-

→ Truth ^{value} of the proposition (God is one) is T

→ " " " (Birds do not fly) is F.

→ Truth value of proposition ($x+2=10$) and (Shut the door) is None (as it is not a proposition)

Conditional related with other Conditionals:-

Let $P \rightarrow Q$ be a given conditional, then

i- $Q \rightarrow P$ is called the **Converse** of $P \rightarrow Q$.

ii- $\sim P \rightarrow \sim Q$ is called the **Inverse** of $P \rightarrow Q$.

iii- $\sim Q \rightarrow \sim P$ is called the **Contra positive** of $P \rightarrow Q$.

Truth table for (i), (ii) and (iii) is

				Conditional	Converse	Inverse	Contra-Positive
P	Q	$\sim P$	$\sim Q$	$P \rightarrow Q$	$Q \rightarrow P$	$\sim P \rightarrow \sim Q$	$\sim Q \rightarrow \sim P$
T	T	F	F	T	T	T	T
T	F	F	T	F	T	T	F
F	T	T	F	T	F	F	T
F	F	T	T	T	T	T	T

From this table, it is clear that

→ Any conditional and its contrapositive are equivalent. Therefore, any theorem can be proved by proving its contrapositive.

→ The converse and Inverse are also equivalent.

→ Tautology:-

A compound (statement) proposition which is always true for all values of the variables involved, is called tautology.

Examples:-

P or $\sim P$, symbolically $P \vee \sim P$ is always True, so, it is tautology.

→ Contradiction:-

A compound proposition (statement) which is always false for all values of the variables involved, is called Contradiction or Absurdity.

Ex#

P and $\sim P$, i.e. $(P \wedge \sim P)$ is always false. So, it is contradiction or Absurdity.

→ Contingency:-

A compound proposition which is neither true nor false is called a contingency.

Ex#

$P \rightarrow \sim P$ is neither true nor false. So,

Date: _____

Day: _____

it is contingency.

Proposition	Negation	tautology	Contradiction	Contingency
P	$\sim P$	$P \vee \sim P$	$P \wedge \sim P$	$P \rightarrow \sim P$
T	F	T	F	F
F	T	T	F	T

Exercise:-

Ex #01:-

Write converse, inverse and contrapositive of the following conditional $\sim p \rightarrow q$

Sol:-

Conditional	converse	Inverse	Contrapositive
$\sim p \rightarrow q$	$q \rightarrow \sim p$	$p \rightarrow \sim q$	$\sim q \rightarrow p$

Ex #02:-

Construct truth table for $\sim(p \rightarrow q) \leftrightarrow (p \wedge \sim q)$

P	q	$\sim q$	$P \rightarrow q$	$\sim(P \rightarrow q)$	$(P \wedge \sim q)$	
T	T	F	T	F	F	T
T	F	T	F	T	T	T
F	T	F	T	F	F	T
F	F	T	T	F	F	T

"Logical Equivalences"

Def:-

The compound propositions p and q are said to be logically equivalent if $p \leftrightarrow q$ is a tautology. Logical equivalence is denoted by $\equiv$ or $\Leftrightarrow$.

Ex#

$$p \wedge T \equiv p$$

P	T	$p \wedge T$
T	T	T
F	T	F

Most common Logical Equivalences:-

i) Identity Laws:-

(a) $p \wedge T \equiv p$

(b) $p \vee F \equiv p$

ii) Domination laws:-

(a) $p \vee T \equiv T$

(b) $p \wedge F \equiv F$

iii) Idempotent Laws:-

(a) $p \vee p \equiv p$

(b) $p \wedge p \equiv p$

iv) Double Negation Law:-

(a) $\sim(\sim p) \equiv p$

v) Commutative Laws:-

(a) $p \vee q \equiv q \vee p$

(b) $p \wedge q \equiv q \wedge p$

Date: _____

Day: _____

vi) Associative laws:-

$$(a) (P \vee Q) \vee R \equiv P \vee (Q \vee R)$$

$$(b) (P \wedge Q) \wedge R \equiv P \wedge (Q \wedge R)$$

vii) Distributive laws:-

$$(a) P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)$$

$$(b) P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$$

viii) De Morgan's laws:-

$$(a) \sim(P \wedge Q) \equiv \sim P \vee \sim Q$$

$$(b) \sim(P \vee Q) \equiv \sim P \wedge \sim Q$$

ix) Absorption laws:-

$$(a) P \vee (P \wedge Q) \equiv P$$

$$(b) P \wedge (P \vee Q) \equiv P$$

x) Negation laws:-

$$(a) P \vee \sim P \equiv T$$

$$(b) P \wedge \sim P \equiv F$$

logical equivalences involving biconditionals

$$i) P \leftrightarrow Q \equiv (P \rightarrow Q) \wedge (Q \rightarrow P)$$

$$ii) P \leftrightarrow Q \equiv \sim P \leftrightarrow \sim Q$$

$$iii) P \leftrightarrow Q \equiv (P \wedge Q) \vee (\sim P \wedge \sim Q)$$

$$iv) \sim(P \leftrightarrow Q) \equiv P \leftrightarrow \sim Q$$

logical eq. involving conditionals

$$i) P \rightarrow Q \equiv \sim P \vee Q$$

$$ii) P \rightarrow Q \equiv \sim Q \rightarrow \sim P$$

Date: _____

Day: _____

- iii) $\sim(P \rightarrow q) \equiv P \wedge \sim q$
- iv) $P \vee q \equiv \sim P \rightarrow q$
- v) $P \wedge q \equiv \sim(q \rightarrow \sim P)$
- vi) $(P \rightarrow q) \wedge (P \rightarrow r) \equiv P \rightarrow (q \wedge r)$
- vii) $(P \rightarrow q) \vee (P \rightarrow r) \equiv P \rightarrow (q \vee r)$
- viii) $(P \rightarrow r) \vee (q \rightarrow r) \equiv (P \wedge q) \rightarrow r$
- ix) $(P \rightarrow r) \wedge (q \rightarrow r) \equiv (P \vee q) \rightarrow r$

"Ven Diagram"

→ Set:-

Cardinal of a set

$$A = \{a, b, c, d\}$$

It is denoted by $|A|$.

$$\text{So, } |A| = 4; [A] = 4.$$

$$A \cup B = \{x: x \in A \text{ or } x \in B\}$$

$$A \cap B = \{x: x \in A \text{ and } x \in B\}$$

$$A - B = \{x: x \in A \text{ but } x \notin B\}$$

$$B - A = \{x: x \in B \text{ but } x \notin A\}$$

$$A' = U - A = \{x: x \in U \text{ but } x \notin A\}$$

→ Laws

$$A \cup A = A; A \cap A = A$$

$$A \cup \phi = A; A \cap U = A$$

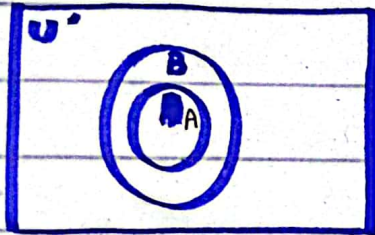
$$\phi' = U; U' = \phi$$

$$A \cup (A \cap B) = A$$

$$A \cap (A \cup B) = A$$

$$A - B = A \cap B'$$

i) $A \subseteq B$



$$A \cup B = B$$

$$A \cap B = A$$

$$A - B = \phi$$

$$B - A = ??$$

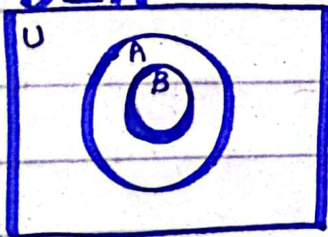
$$n(A \cup B) = n(B)$$

$$n(A \cap B) = n(A)$$

$$n(A - B) = 0$$

$$n(B - A) = n(B) - n(A)$$

ii) $B \subseteq A$



$$A \cup B = A$$

$$A \cap B = B$$

$$A - B = ??$$

$$B - A = \phi$$

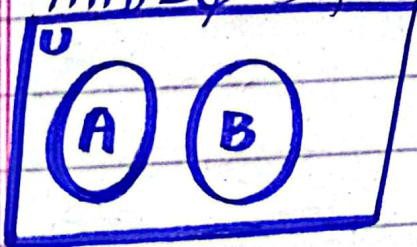
$$n(A \cup B) = n(A)$$

$$n(A \cap B) = n(B)$$

$$n(A - B) = n(A) - n(B)$$

$$n(B - A) = 0$$

iii) $A \cap B = \phi$ Disjoint



$$A \cup B = ?$$

$$A \cap B = \phi$$

$$A - B = A$$

$$B - A = B$$

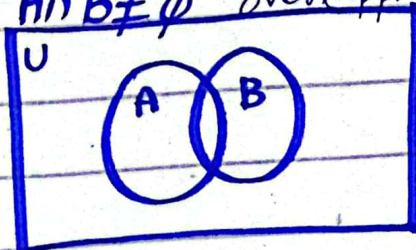
$$n(A \cup B) = n(A) + n(B)$$

$$n(A \cap B) = 0$$

$$n(A - B) = n(A)$$

$$n(B - A) = n(B)$$

iv) $A \cap B \neq \phi$ overlapping



$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$n(A - B) = n(A) - n(A \cap B)$$

$$n(B - A) = n(B) - n(A \cap B)$$

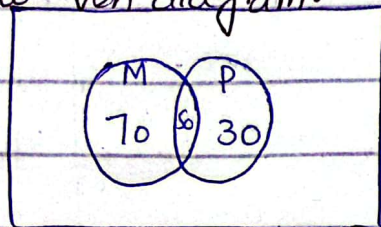
for ex#

A Maths 120 students

B Physics 80 students

Both Math and Physics 50 students.

Draw ven diagram.



$$|A \cup B| = |A| + |B| - |A \cap B|$$

$$= 120 + 80 - 50$$

$$= 150$$

So, $|A \cup B|^{150}$ represents the total no. of students studying either maths or physics or both.

Ex#2 languages No. of learners

French 65

$$F + G = 20$$

German 45

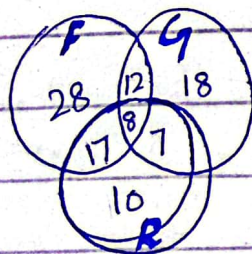
$$F + R = 25$$

Russian 42

$$G + R = 15$$

Draw ven diagram.

$$F + G + R = 8$$



Now, let's calculate the total no. of students studying at least one of the

three languages

$$|F \cup G \cup R| = |F| + |G| + |R| - |FG| - |FR| - |GR| + |FGR|$$

$$= 65 + 45 + 42 - 20 - 25 - 15 + 8$$

$$= 100$$

So,

$|F \cup G \cup R| = 100$. This represents the total no.'s of students studying at least one of French, German or Russian.